Network Security Documentation

ISEAGE International Cyber Defense Competition 2026

Team 37 — CtrlAltDefend | University of Mitrovica "Isa Boletini"

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1. Executive Summary

Team CtrlAltDefend is tasked with defending the Central Downtown City (CDC) Public Transit System infrastructure. Our network consists of five servers supporting ticketing, administration, public information, and backend management services for the city's bus and train operations.

Our defense strategy follows a **Defense-in-Depth** approach with four layers: 1. **Perimeter defense** — Host-based firewalls with explicit allow rules 2. **System hardening** — Removal of backdoors, malicious users, and unnecessary services 3. **Application security** — Secure configurations, patched code, strong session management 4. **Monitoring & response** — Real-time log monitoring, anomaly detection, and documented incident response

We employ an **centralized command center** for real-time coordination across all five servers, enabling rapid threat detection and strategic decision-making.

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2. Team Organization

Member	Role	Assigned Systems	Expertise
Agon Bajgora	Team Leader & Team Coordinator	Team coordination & strategy	10+ years software development
Valdrin Bislimi	Windows Infrastructure Lead	AD Server, Ticketing Server	Windows Server administration, Active Directory
Flutura Hyseni	Windows Application Specialist	Admin UI Server	Windows Server, web application management
Erza Osmani	Linux Backend Specialist	Management Backend	Linux administration, database management
G Voca	Linux Web Specialist & Documentation	Public Website	Linux administration, web servers, technical writing

Communication Protocol

- **Primary:** Team chat group chat with team leader for real-time command support
- **Escalation:** Critical security incidents reported immediately to Team Leader
- **Documentation:** G Voca maintains running incident log; all members report changes

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3. Network Architecture

3.1 Network Layout

Network Segment: 10.0.37.0/24 (Team 37 internal network)
NAT: Not implemented — direct routing as provided by competition infrastructure
External DNS: *.team37.isucdc.com (managed by competition)

3.2 Network Diagram

```
||||||||||||||||||
|    INTERNET      |
|  (Red Team /     |
|   Green Team)    |
||||||||||||||||||
          |
||||||||||||||||||
|  COMPETITION    |
|  GATEWAY        |
||||||||||||||||||
          |
Team 37 Network: 10.0.37.0/24
```

```
                                |
          |||||||||||||||||||||||||||||||||||||||||||||||
          |          |            |            |          |
     ||||||||||| |||||||||| |||||||||| |||||||||||| ||||||||||||
     |   AD    | |TICKETS | | ADMIN  | |  MGMT   | |  WWW    |
     |         | |        | |   UI   | | BACKEND | | PUBLIC  |
     |Win 2016 | |Win 2019| |Win 2022| |Alma 8.4 | |Alma 9   |
     |  .10    | |  .20   | |  .30   | |  .40    | |  .50    |
     |         | |        | |        | |         | |         |
     |LDAP:389 | |HTTP:   | |HTTP:80 | |HTTP:    | |HTTP:80  |
     |LDAPS:636| |  5000  | |RDP:    | |  5000   | |SSH:22   |
     |DNS:53   | |RDP:    | |  3389  | |SSH:22   | |         |
     |Kerb:88  | |  3389  | |        | |         | |         |
     |RDP:3389 | |        | |        | |         | |         |
     ||||||||||| |||||||||| |||||||||| |||||||||||| ||||||||||||
          |                      |            |
          |    ||||||||||||||||||            |
          |    |  ADMIN UI queries           |
          |    |  MGMT backend API ||||||||||||
          |    |  (BACKEND_URL=mgmt.team37.isucdc.com:5000)
          |
          ||| All servers authenticate against AD (LDAP/Kerberos)
```

3.3 Service Dependency Map

```
||||||||||||||||||||||||||||||||||||||||||||||||||
|                 ACTIVE DIRECTORY                 |
|          Central Authentication (LDAP)           |
|    All domain users authenticate here first      |
||||||||||||||||||||||||||||||||||||||||||||||||||
       |          |          |          |
       |          |          |          |
    |||||||||  |||||||||  |||||||||  ||||||||||
    | WWW   |  | MGMT  |  |ADMIN  |  |TICKETS |
    |       |  |       |  |   |   |  |        |
    |Login  |  |SSH    |  |Login  |  |Auth    |
    |& SSH  |  |Auth   |  |Auth   |  |        |
    |||||||||  |||||||||  |||||||||  ||||||||||
                 |          |
                 ||||||||||||
               ADMIN calls MGMT
               API for transit data
```

3.4 DNS Resolution

Hostname	IP Address	Service
ad.team37.isucdc.com	10.0.37.10	Active Directory / DNS

Hostname	IP Address	Service
tickets.team37.isucdc.com	10.0.37.20	Ticketing API
admin.team37.isucdc.com	10.0.37.30	Admin Web UI
mgmt.team37.isucdc.com	10.0.37.40	Management Backend API
www.team37.isucdc.com	10.0.37.50	Public Website

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4. Server Inventory & Operating Systems

4.1 AD Server (Team-37-AD)

Property	Value
OS	Windows Server 2016
IP	10.0.37.10
Function	Active Directory Domain Controller, DNS Server
Services	LDAP (389), LDAPS (636), DNS (53), Kerberos (88), RDP (3389)
Software	Active Directory Domain Services, DNS Server role
Domain Users	10 accounts (as provided, all retained per rules)
Managed by	Valdrin Bislimi

4.2 Ticketing Server (Team-37-TICKETS)

Property	Value
OS	Windows Server 2019
IP	10.0.37.20
Function	Customer Ticketing API
Services	HTTP (5000), RDP (3389)
Software	.NET 8 Runtime, ASP.NET Web API
Application Path	`C:\Users\Administrator\Desktop\tickets\`
Cert Thumbprint	`4ee49076b7df8b39ad292ae48d5809dad45876a9`
Managed by	Valdrin Bislimi

4.3 Admin UI Server (Team-37-ADMIN)

Property	Value
OS	Windows Server 2022
IP	10.0.37.30
Function	Administrative Web Interface
Services	HTTP (80), RDP (3389)

Property	Value
Software	Python, Flask Web Framework
Application Path	`C:\Users\Administrator\Desktop\icdc26-admin-u`
Backend Connection	`http://mgmt.team37.isucdc.com:5000`
Managed by	Flutura Hyseni

4.4 Management Backend (Team-37-MGMT)

Property	Value
OS	AlmaLinux 8.4 (RHEL-based)
IP	10.0.37.40
Function	Backend REST API + Database
Services	HTTP (5000), SSH (22)
Software	Python, Flask, MariaDB
Application Path	`/home/cdc/icdc26-backend-mgmt/`
Managed by	Erza Osmani

4.5 Public Website (Team-37-WWW)

Property	Value
OS	AlmaLinux 9 (RHEL-based)
IP	10.0.37.50
Function	Public-Facing Transit Information Portal
Services	HTTP (80), SSH (22)
Software	Python, Flask Web Framework
Managed by	G Voca

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5. Security Hardening Plan

5.1 Phased Approach

Our hardening follows a strict priority order to minimize exposure time:

Phase 1 — Critical (0-15 minutes):

Priority	Action	Server	Responsible
1	Change all default "cdc" passwords	ALL	Everyone
2	Change AD domain user passwords	AD	Valdrin
3	Remove attacker user `taco`	WWW	G Voca
4	Remove attacker users `nsa`, `scrat`	ADMIN	Flutura
5	Remove attacker users `backdoor`, `scrat`	TICKETS	Valdrin
6	Remove Guest from Administrators group	ADMIN	Flutura

Phase 2 — Infrastructure (15-30 minutes):

Priority	Action	Server	Responsible
7	Kill `odd_jobd.service` persistence	WWW, MGMT	G Voca, Erza
8	Kill netcat backdoors (1337, 12345, 12869)	WWW	G Voca
9	Enable host firewalls (see Section 6)	ALL	Everyone
10	Restore Windows Defender	ADMIN	Flutura
11	Start Ticketing application	TICKETS	Valdrin

Phase 3 — Application (30-60 minutes):

Priority	Action	Server	Responsible
12	Change Flask SECRET_KEY from "cdc"	WWW, ADMIN	G Voca, Flutura
13	Set HTTPONLY=True on cookies	WWW, ADMIN	G Voca, Flutura
14	Verify all scored services responding	ALL	Agon (via curl)
15	Place blue flags	ALL	Everyone

Phase 4 — Ongoing Defense (60+ minutes):

Priority	Action	Server	Responsible
16	Monitor logs for intrusion	ALL	Everyone
17	Guard red flag directories	ALL	Everyone
18	Attempt special flags (forged ticket, DB)	TICKETS, MGMT	Valdrin, Erza
19	Document all incidents	—	G Voca

5.2 Password Policy

Parameter	Value
Minimum length	15 characters
Complexity	Uppercase + lowercase + numbers + symbols
Domain password	Shared across team (Valdrin sets on AD)
Local admin passwords	Unique per server, per team member

Parameter	Value
Application secrets	Unique random strings per application
Rotation	Every 2 hours during competition if breach suspected

5.3 Pre-Existing Threats Identified

Threat	Server	Severity	Remediation
Netcat backdoors on ports 1337, 12345, 12869	WWW	**CRITICAL**	Kill processes, block ports
`odd_jobd.service` persistence mechanism	WWW, MGMT	**CRITICAL**	Stop, disable, delete unit file
Attacker user `taco`	WWW	**HIGH**	Delete user, remove home dir
Attacker users `nsa`, `scrat` with admin	ADMIN	**HIGH**	Delete users
Attacker users `backdoor`, `scrat` with admin	TICKETS	**HIGH**	Delete users
Guest account in Administrators group	ADMIN	**HIGH**	Remove from group
Windows Defender completely removed	ADMIN	**HIGH**	Reinstall/enable
All firewalls disabled	WWW, MGMT, ADMIN	**HIGH**	Enable with strict rules
Flask SECRET_KEY = "cdc"	WWW, ADMIN	**MEDIUM**	Change to random string
HTTPONLY = False on session cookies	WWW, ADMIN	**MEDIUM**	Set to True
SSH logins from suspicious accounts	MGMT	**MEDIUM**	Audit and lock accounts

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6. Firewall Policy & Rules

6.1 General Policy

- **Default policy:** DENY ALL inbound
- **Strategy:** Explicit allow only for required service ports
- **Outbound:** Allow all (needed for AD authentication and inter-server communication)

6.2 Linux Firewall Rules (firewalld)

```
firewall-cmd --set-default-zone=drop
firewall-cmd --permanent --add-port=80/tcp    # HTTP (scored)
firewall-cmd --permanent --add-port=22/tcp    # SSH (scored)
firewall-cmd --reload
```

WWW Server (AlmaLinux 9)

```
firewall-cmd --set-default-zone=drop
firewall-cmd --permanent --add-port=5000/tcp  # API (scored)
firewall-cmd --permanent --add-port=22/tcp    # SSH (scored)
firewall-cmd --reload
```

MGMT Server (AlmaLinux 8.4)

6.3 Windows Firewall Rules (PowerShell)

```
# Enable all profiles
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True -DefaultInboundAction Bloc
# Allow required services
New-NetFirewallRule -DisplayName "LDAP" -Direction Inbound -Port 389 -Protocol TCP -Action All
New-NetFirewallRule -DisplayName "LDAPS" -Direction Inbound -Port 636 -Protocol TCP -Action Al
New-NetFirewallRule -DisplayName "DNS-TCP" -Direction Inbound -Port 53 -Protocol TCP -Action A
New-NetFirewallRule -DisplayName "DNS-UDP" -Direction Inbound -Port 53 -Protocol UDP -Action A
New-NetFirewallRule -DisplayName "Kerberos" -Direction Inbound -Port 88 -Protocol TCP -Action
New-NetFirewallRule -DisplayName "RDP" -Direction Inbound -Port 3389 -Protocol TCP -Action All
```

AD Server

```
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True -DefaultInboundAction Bloc
New-NetFirewallRule -DisplayName "TicketAPI" -Direction Inbound -Port 5000 -Protocol TCP -Acti
New-NetFirewallRule -DisplayName "RDP" -Direction Inbound -Port 3389 -Protocol TCP -Action All
```

TICKETS Server

```
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True -DefaultInboundAction Bloc
New-NetFirewallRule -DisplayName "AdminHTTP" -Direction Inbound -Port 80 -Protocol TCP -Action
New-NetFirewallRule -DisplayName "RDP" -Direction Inbound -Port 3389 -Protocol TCP -Action All
```

ADMIN Server

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7. Intrusion Detection & Monitoring

7.1 Log Monitoring Strategy

Server	Log Source	What We Monitor
WWW	`/var/log/secure`, `/var/log/messages`	Failed SSH, new users, sudo attempts
MGMT	`/var/log/secure`, `/var/log/messages`, MariaDB logs	Failed logins, DB queries, API errors
AD	Windows Security Event Log	Failed auth (4625), account changes (4720/4722), privilege escalation
TICKETS	Windows Security Event Log	Failed logins, new accounts, service changes
ADMIN	Windows Security Event Log	Failed logins, new accounts, Defender alerts

7.2 Real-Time Monitoring Commands

Linux (run in background):

```
# Watch for authentication failures
tail -f /var/log/secure | grep -i "failed\|invalid\|error"

# Watch for new processes
watch -n 5 'ps aux --sort=-%cpu | head -20'

# Monitor network connections
watch -n 10 'ss -tunap | grep -v LISTEN'
```

Windows (PowerShell):

```
# Watch for failed logins (Event ID 4625)
Get-WinEvent -FilterHashtable @{LogName='Security';ID=4625} -MaxEvents 10

# Check for new user accounts (Event ID 4720)
Get-WinEvent -FilterHashtable @{LogName='Security';ID=4720} -MaxEvents 10

# Active connections
Get-NetTCPConnection | Where-Object {$_.State -eq 'Established'}
```

7.3 File Integrity Monitoring

Blue Flag Protection:

```
# Linux: Monitor flag directories
inotifywait -m /etc/ -e create,modify,delete 2>/dev/null &
inotifywait -m /root/ -e create,modify,delete 2>/dev/null &
```

```
# Windows: Check flag directory
$watcher = New-Object System.IO.FileSystemWatcher 'C:\Windows\System32'
$watcher.EnableRaisingEvents = $true
```

7.4 Anomaly Detection Indicators

Indicator	Action
New user account created	Investigate immediately, delete if unauthorized
New listening port	Identify process, kill if unauthorized
Outbound connection to unknown IP	Block IP, investigate process
Modified system binary	Restore from known good, investigate
cron/scheduled task added	Review and remove if malicious
Flag file modified or deleted	Restore immediately

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8. Access Control & Authentication

8.1 Active Directory Policy

Setting	Value
Password minimum length	15 characters
Account lockout threshold	5 failed attempts
Account lockout duration	30 minutes
Password complexity	Enabled
Audit logon events	Success and Failure
Anonymous enumeration	Disabled

8.2 Linux Access Controls

Setting	Value
Root SSH login	Disabled (`PermitRootLogin no`)
SSH password auth	Enabled (domain users need it)
Sudo access	Restricted to authorized users only
File permissions on flags	644 (root owned)
umask	027

8.3 Mandatory Access Controls

- **SELinux:** Enforcing mode on AlmaLinux servers (default, maintained)
- **Windows UAC:** Enabled on all Windows servers

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9. Application Security

9.1 Flask Applications (WWW, ADMIN, MGMT)

Fix	Before	After
SECRET_KEY	"cdc" (default)	Random 64-character string
SESSION_COOKIE_HTTPONLY	False	True
SESSION_COOKIE_SECURE	False	True (if HTTPS available)
DEBUG mode	Potentially enabled	Disabled

9.2 .NET Ticketing Application (TICKETS)

Configuration	Value
Signing certificate	Thumbprint verified and set
HTTPS	Configured if cert available
Error pages	Custom (no stack traces)
Input validation	Framework defaults + review

9.3 Database Security (MGMT — MariaDB)

Action	Detail
Root password	Changed from default
Remote root login	Disabled
Anonymous users	Removed
Test database	Removed
Application DB user	Minimal privileges (SELECT, INSERT, UPDATE only)

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10. Incident Response Plan

10.1 Severity Levels

Level	Description	Response Time	Example
P1 — Critical	Service down or flag compromised	Immediate	Scored service offline, red flag planted
P2 — High	Active intrusion detected	$<$ 5 minutes	New unauthorized user, backdoor found
P3 — Medium	Suspicious activity	$<$ 15 minutes	Brute force attempts, port scan
P4 — Low	Anomaly detected	$<$ 30 minutes	Unusual log entry

10.2 Response Procedure

`DETECT -> REPORT -> CONTAIN -> ERADICATE -> RECOVER -> DOCUMENT`

1. **Detect:** Identify the incident via logs, monitoring, or scoring alerts
2. **Report:** Report to team with server name, severity, and description
3. **Contain:** Isolate the threat (kill process, block IP, disable account)
4. **Eradicate:** Remove all traces (backdoors, malware, unauthorized accounts)
5. **Recover:** Restore services, verify scoring checks pass
6. **Document:** G Voca logs the incident with timestamp and resolution

10.3 Emergency Contacts

- **Team Leader (Agon):** Available at all times
- **Competition Organizer:** Rrezearta Thaqi (+383 45 686 169)

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11. Backup & Recovery

11.1 Configuration Backups

Each team member creates backups of critical configs immediately after hardening:

Linux:

```
cp /etc/ssh/sshd_config /etc/ssh/sshd_config.bak
cp /etc/firewalld/ /root/firewall-backup/ -r
```

Windows:

```
# Export firewall rules
netsh advfirewall export C:\firewall-backup.wfw
# Export group policy
gpresult /H C:\gpo-backup.html
```

11.2 Application Backups

- Copy application directories before any code changes
- Database dump: `mysqldump --all-databases > /root/db-backup.sql`

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12. Appendix

A. Scored Service Verification

Service	Check Command	Expected Result
WWW HTTP	`curl http://www.team37.isucdc.com`	HTTP 200, HTML page
WWW SSH	`ssh cdc@www.team37.isucdc.com`	Login prompt

Service	Check Command	Expected Result
ADMIN Login	`curl http://admin.team37.isucdc.com`	HTTP 200, login page
TICKETS API	`curl http://tickets.team37.isucdc.com:5000`	HTTP 200, API response
MGMT API	`curl http://mgmt.team37.isucdc.com:5000`	HTTP 200, API response
AD DNS	`nslookup www.team37.isucdc.com 10.0.37.10`	Resolves correctly

B. Special Flag Targets

Flag	Server	Method
Forged Ticket	TICKETS	Submit ticket with `validUntil >= 2000000000`
DB Read	MGMT	Query MariaDB for flag data
Bus/Train Service	MGMT	Interact with transit service via API

C. Blue Flag Locations

Server	OS	Blue Flag Path
WWW	Linux	`/etc/`
MGMT	Linux	`/etc/`
AD	Windows	`C:\Windows\System32\`
TICKETS	Windows	`C:\Windows\System32\`
ADMIN	Windows	`C:\Windows\System32\`

D. Red Flag Directories (Defend)

Server	OS	Red Flag Path
WWW	Linux	`/root/`
MGMT	Linux	`/root/`
AD	Windows	`C:\Users\Administrator\`
TICKETS	Windows	`C:\Users\Administrator\`
ADMIN	Windows	`C:\Users\Administrator\`

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This document represents our comprehensive network security plan for the CDC Transit System defense. All measures described are implemented at competition start with the goal of maintaining maximum service availability while preventing unauthorized access.